C Programming - Assignment 08

Q.1 Write Functions to Accept and Print One Student Record

Problem Statement

Write a program to declare a structure to store student information. The structure should contain Roll Number, Name, and Marks. Write separate functions to accept and print one student record.

Concepts Used

- Structure
- Functions
- Pointers to Structure
- User Input / Output
- Structure Members

Step-by-Step Approach

1. Define a structure named `Student`.
2. Add members: Roll Number, Name, and Marks.
3. Create a function to accept student details.
4. Create another function to display student details.
5. Declare a structure variable in `main()`.
6. Call the accept function to store data.
7. Call the print function to display the record.

Test Cases

Test Case	Roll No	Name	Marks
1	101	Amit	85.50
2	102	Neha	90.00
3	103	Raj	78.50
4	104	Priya	88.00
5	105	Kunal	92.50

Expected Output

```
Roll No : 101
Name    : Amit
Marks   : 85.50
```

Q.2 Accept and Print Array of Student Structures

Problem Statement

Write functions to accept and print an array of student structures.

Concepts Used

- Structure
- Array of Structures
- Functions
- Looping (`for`)
- User Input / Output
- Structure Members

Step-by-Step Approach

1. Define a structure named `Student`.
2. Create an array of student structures.
3. Write a function to accept details of all students.
4. Use a loop to store multiple student records.
5. Write another function to display all student records.
6. Use a loop to print all student records.
7. Call both functions from `main()`.

Test Cases

Test Case 1

Roll No	Name	Marks
101	Amit	85.50
102	Neha	90.00
103	Raj	78.50

Expected Output

```
Roll No : 101
Name    : Amit
Marks   : 85.50

Roll No : 102
Name    : Neha
Marks   : 90.00

Roll No : 103
```

```
Name     : Raj
Marks    : 78.50
```

Test Case 2

Roll No	Name	Marks
201	Pooja	82.00
202	Rohan	76.50
203	Sneha	91.00

Expected Output

```
Roll No : 201
Name     : Pooja
Marks    : 82.00

Roll No : 202
Name     : Rohan
Marks    : 76.50

Roll No : 203
Name     : Sneha
Marks    : 91.00
```